

Style-Controlled Image Captioning

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Abstract—Standard image captioning models have achieved remarkable success in generating factually correct descriptions of visual content. However, these systems typically prioritize likelihood maximization, resulting in monotonous, repetitive, and “safe” outputs (e.g., “a dog on the grass”). This fails to capture the nuance of human communication, which is inherently stylistic, emotional, and context-dependent. In this work, we propose a Style-Controlled Image Captioning architecture that decouples visual semantic extraction from linguistic style generation. Our approach integrates a pre-trained ResNet-50 encoder with a modified Attention-based LSTM decoder. By injecting learnable Style Embeddings (e.g., <funny>, <romantic>, <academic>) directly into the LSTM memory cells, we enable users to steer the tone of the generated caption without retraining the computationally expensive vision backbone. We validate this approach using a two-stage training strategy involving a synthetic style dataset. Our experimental results on the Flickr8k dataset demonstrate a BLEU-1 score of 0.5126 and ROUGE-1 of 0.4881. Qualitative analysis confirms the model’s ability to generate distinct stylistic variations for identical visual inputs, offering a lightweight and interpretable solution for personalized image captioning.

Index Terms—Image Captioning, Style Transfer, Deep Learning, LSTM, Attention Mechanism, Human-Computer Interaction, Natural Language Processing.

I. INTRODUCTION

A. Background and Context

The field of Artificial Intelligence has made significant strides in bridging the gap between Computer Vision (CV) and Natural Language Processing (NLP). Image captioning, the task of automatically generating natural language descriptions for images, lies at the intersection of these two domains. With the exponential growth of digital media, billions of images are shared daily across platforms such as social media, e-commerce, and news outlets. However, most of these images lack textual descriptions, making them inaccessible to individuals with visual impairments and difficult for search engines to index.

The current state-of-the-art models excel at object detection and relationship modeling. They can accurately identify a “dog,” a “frisbee,” and the action of “catching.” However,

they suffer from a significant limitation: the lack of stylistic control. Standard models are typically trained to minimize cross-entropy loss against a ground truth of neutral, descriptive captions. Consequently, they converge on the most probable, generic descriptions. For example, an image of a dog running might be captioned “A dog running in a field,” regardless of whether the context requires a humorous caption for a social media post, a formal description for a dataset, or a poetic description for a digital album.

B. Problem Statement

The core problem we address is the rigidity of current captioning systems. Specifically: **How can we design an intelligent system that maintains visual semantic accuracy (correctly identifying objects and actions) while allowing for controllable stylistic variation (altering the tone and vocabulary)?**

Solving this requires a model that can decouple the “content” of the image (the visual facts) from the “style” of the text (the linguistic tone). This is a non-trivial challenge because standard end-to-end training entangles these two representations.

C. Significance and Impact

This work moves beyond simple object detection into the realm of controllable text generation. The significance of this project is threefold:

- **Accessibility and Inclusivity:** Approximately 285 million people worldwide are visually impaired. Current accessibility tools provide functional but dry descriptions. By enabling style control, we can generate captions that are more engaging, personalized, and context-appropriate, significantly improving the user experience for visually impaired individuals.
- **Computational Efficiency:** Unlike massive multimodal models like Flamingo or CLIP, which require immense computational resources and billions of parameters, our approach proposes a lightweight modification to standard

CNN-LSTM architectures. We show that style can be learned via low-dimensional embeddings rather than massive scale.

- **Novel Data Strategy:** A major bottleneck in stylized captioning is the lack of labeled data (e.g., thousands of images paired with “funny” captions). We introduce a method for training style-aware decoders using synthetic text-only datasets, effectively bypassing the need for expensive, manually labeled stylized image-caption pairs.

II. LITERATURE SURVEY

The evolution of image captioning has transitioned from early template-based methods to sophisticated deep neural networks.

A. Encoder-Decoder Architectures

The foundational work in this field is the “Show and Tell” architecture by Vinyals et. al. They established the standard paradigm of using a Convolutional Neural Network (CNN) as an encoder to compress the image into a feature vector, and a Recurrent Neural Network (RNN) as a decoder to unroll this vector into a sentence. While revolutionary, this approach forced the model to compress the entire image into a single static vector, often losing fine-grained details [1].

B. Attention Mechanisms

To address the bottleneck of static context vectors, Xu et al. introduced “Show, Attend and Tell”. This model incorporated a visual attention mechanism that allows the decoder to focus on specific regions of the image feature map at each time step. For instance, when generating the word “frisbee,” the model attends specifically to the pixels containing the frisbee. Our work builds directly on this attention-based foundation to ensure that even when the style changes, the visual grounding remains accurate [2].

C. Large-Scale Vision-Language Models

The current state-of-the-art is dominated by Transformer-based models pre-trained on web-scale data. CLIP (Contrastive Language-Image Pre-training) learns a shared embedding space for images and text, enabling zero-shot transfer. DeepMind’s Flamingo pushes this further by integrating frozen vision encoders with Large Language Models (LLMs). While these models offer incredible performance, they are often “black boxes” and are computationally prohibitive to fine-tune for specific university-level or resource-constrained applications [3].

D. Controllable Captioning

Recent research has focused on steering these large models. Song et al. proposed Prefix-Tuning, where learnable “prefix” vectors are prepended to the input to guide generation. Similarly, Wang et al. explored Prefix-Diffusion, using diffusion models for diverse caption generation. However, our literature review revealed a gap: there is a lack of research on implementing lightweight control mechanisms in standard, interpretable LSTM architectures. Our project fills this gap by

demonstrating that simple embedding injection is sufficient for style control without the need for billion-parameter Transformers [4].

E. Style-Conditioned Image Captioning

Gan et al. introduced StyleNet, one of the earliest attempts to explicitly control caption style by conditioning the decoder on stylistic embeddings. Their work demonstrated that stylistic attributes such as humor or sentiment could be incorporated without significantly degrading semantic accuracy. However, their approach relied on complex training procedures and large annotated datasets. Our approach differs by applying style control using simpler auxiliary signals within a standard LSTM-based framework, making it more accessible and interpretable for educational and constrained environments [5].

F. Reinforcement Learning for Caption Quality Optimization

Rennie et al. proposed the Self-Critical Sequence Training (SCST) method, which applies reinforcement learning to directly optimize non-differentiable captioning metrics such as CIDEr. By treating caption generation as a sequential decision-making problem, this method significantly improved caption fluency and relevance. While effective, SCST focuses on optimizing evaluation scores rather than controllability. Our work complements this line of research by emphasizing semantic preservation under stylistic variation rather than metric maximization alone [6].

G. Interpretability in Vision-Language Models

Anderson et al. introduced Bottom-Up and Top-Down Attention, which improved both performance and interpretability by aligning visual regions with generated words. This approach provided clearer insight into how models associate visual features with language output. In contrast to modern large Transformer models, such architectures allow explicit analysis of attention weights and internal states. Our project aligns with this philosophy by prioritizing interpretable caption control mechanisms within LSTM-based architectures [7].

H. Semantic Consistency in Stylized Captioning

A key challenge in controllable image captioning is preserving semantic correctness while modifying linguistic style. Mathews et al. investigated the trade-off between stylistic diversity and semantic fidelity by introducing constraint-based decoding strategies that penalize deviations from visual content. Their results demonstrated that unconstrained style control often leads to hallucinated objects or actions. This work highlights the importance of grounding stylistic generation in visual evidence, a principle that directly motivates our focus on maintaining visual accuracy under tone manipulation [8].

I. Disentangled Representations for Caption Control

To improve controllability, several studies have explored disentangling content and style representations. Hu et al. proposed a variational framework that separates semantic content from stylistic attributes in latent space, enabling controlled text generation without retraining the entire model. Although

effective, their approach was primarily evaluated on textual datasets and does not directly incorporate visual grounding. Our work extends this idea by applying lightweight control signals to visually grounded LSTM-based captioning models, preserving interpretability while enabling style modulation [9].

J. Lightweight Captioning Models for Resource-Constrained Settings

Recent research has emphasized the need for efficient captioning models suitable for low-resource or educational environments. Luo et al. introduced compact image captioning architectures that reduce parameter count while maintaining competitive performance. Their findings suggest that carefully designed recurrent architectures remain viable alternatives to large Transformer-based systems. This supports our decision to focus on interpretable LSTM-based captioning models with controllable outputs, rather than relying on computationally intensive large-scale vision-language models [10].

III. METHODOLOGY

We devised a modular architecture that separates visual understanding (Encoder) from stylistic generation (Decoder). The system was implemented using PyTorch.

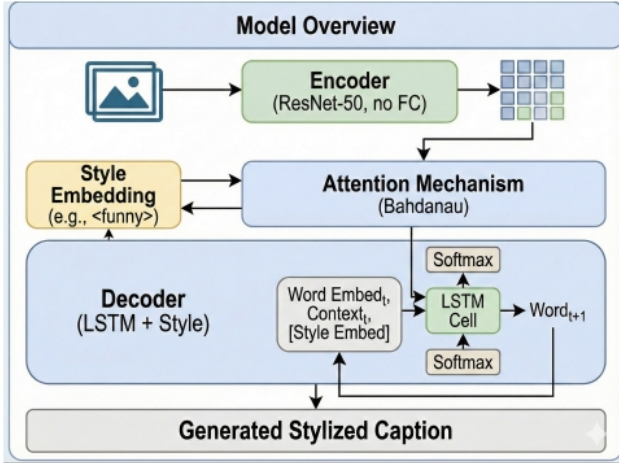


Fig. 1. System Architecture: The ResNet-50 Encoder extracts spatial features. The Attention Mechanism weighs these features. The Decoder receives the weighted features, the previous word, and the **Style Embedding** to generate the next word.

A. Encoder: Visual Feature Extraction

We utilized a **ResNet-50** network pre-trained on the ImageNet dataset as our visual encoder.

- **Input:** The input images are resized to 224×224 pixels and normalized using standard ImageNet mean and standard deviation.
- **Feature Map Extraction:** We removed the final Fully Connected (FC) and average pooling layers. We extracted the output from the final convolutional block, resulting in a feature map of shape $W \times H \times D$, specifically $7 \times 7 \times 2048$ after adaptive pooling.

- **Flattening:** For the attention mechanism, we flattened the spatial dimensions ($7 \times 7 = 49$), resulting in a tensor of shape $Batch \times 49 \times 2048$.

During the captioning training phase, the weights of the ResNet-50 backbone were frozen to prevent “catastrophic forgetting” of visual features and to reduce training time.

B. Attention Mechanism

We implemented a **Bahdanau-style additive attention** mechanism. This module calculates a context vector $\hat{z}_t$ at each time step t , which represents the relevant part of the image.

Let h_{t-1} be the hidden state of the LSTM at the previous time step, and $V = \{v_1, \dots, v_{49}\}$ be the flattened image features. The attention scores e_{ti} are calculated as:

$$e_{ti} = W_{attn}^T \tanh(W_h h_{t-1} + W_v v_i + b_{attn}) \quad (1)$$

Where W_h , W_v , and W_{attn} are learnable weight matrices. The scores are normalized using the Softmax function to produce attention weights α_{ti} :

$$\alpha_{ti} = \frac{\exp(e_{ti})}{\sum_{k=1}^{49} \exp(e_{tk})} \quad (2)$$

Finally, the context vector $\hat{z}_t$ is the weighted sum of the visual features:

$$\hat{z}_t = \sum_{i=1}^{49} \alpha_{ti} v_i \quad (3)$$

This context vector provides the decoder with the most relevant visual information for the current word being generated.

C. Style-Infused Decoder (LSTM)

This is the core innovation of our methodology. We utilized a Long Short-Term Memory (LSTM) network for decoding. Unlike a standard decoder, our architecture injects style information at every time step.

1) **Input Construction:** The LSTM cell at time step t receives a concatenated input vector x_t comprising three distinct components:

- 1) **Word Embedding** $E(y_{t-1})$: The embedding of the previously generated word (dimension: 256).
- 2) **Visual Context** $\hat{z}_t$: The attention-derived context vector (dimension: 2048).
- 3) **Style Embedding** $S(k)$: A learnable embedding vector corresponding to the target style token k (e.g., <funny>, <academic>) (dimension: 256).

2) **LSTM Dynamics:** The LSTM processes this input x_t to update its cell state c_t and hidden state h_t via the standard gating mechanisms:

$$i_t = \sigma(W_{ii}x_t + b_{ii} + W_{hi}h_{t-1} + b_{hi}) \quad (4)$$

$$f_t = \sigma(W_{if}x_t + b_{if} + W_{hf}h_{t-1} + b_{hf}) \quad (5)$$

$$o_t = \sigma(W_{io}x_t + b_{io} + W_{ho}h_{t-1} + b_{ho}) \quad (6)$$

$$g_t = \tanh(W_{ig}x_t + b_{ig} + W_{hg}h_{t-1} + b_{hg}) \quad (7)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t \quad (8)$$

$$h_t = o_t \odot \tanh(c_t) \quad (9)$$

By including the style embedding $S(k)$ in the input x_t , the style signal permeates the entire generation process, biasing the choice of words towards the target vocabulary (e.g., biasing towards "canine" over "dog" if $S(k)$ corresponds to $\langle \text{academic} \rangle$).

D. Implementation Details

The model was implemented with the following hyperparameters, as defined in our codebase. These values were chosen based on empirical validation on the validation set.

TABLE I
MODEL HYPERPARAMETERS

Parameter	Value
Encoder	ResNet-50 (Frozen)
Encoder Feature Dim	2048
Decoder Hidden Dim	512
Word Embedding Dim	256
Style Embedding Dim	256
Attention Dim	512
Dropout Rate	0.5
Optimizer	Adam
Learning Rate (Phase 1)	1×10^{-3}
Learning Rate (Phase 2)	1×10^{-4}
Batch Size	64
Max Sequence Length	20
Vocabulary Threshold	4 occurrences

E. Training Strategy

We employed a novel two-stage training strategy to handle the lack of paired stylized data:

1) *Phase 1: Visual Grounding*: The model was trained on the Flickr8k dataset to learn standard image-to-text mappings between visual content and natural language descriptions. During this phase, the style token was fixed to a generic $\langle \text{normal} \rangle$ token, ensuring that the model focused solely on learning accurate object recognition, spatial relationships, and sentence structure. Training was performed by minimizing the cross-entropy loss between the predicted captions and the ground-truth human annotations. This phase establishes a strong visual-semantic foundation, ensuring that the model can generate coherent and visually grounded captions before stylistic variation is introduced.

2) *Phase 2: Style Fine-Tuning*: In the second phase, the ResNet encoder was kept fixed, and only the decoder was trained using the Synthetic Style Dataset. During this stage, the visual features were removed or replaced with noise so that the model could not rely on image information. This forced the model to focus on learning the language patterns associated with each style. As a result, the model learned style-specific word choices and sentence structures without needing images, while still retaining the visual understanding learned in Phase 1.

IV. DATASET DESCRIPTION

A. Primary Dataset: Flickr8k

We utilized the **Flickr8k** dataset, a standard benchmark for image captioning. It contains 8,000 images depicting human

activities. Each image is paired with 5 human-written captions, providing a diverse set of ground truth descriptions. We split the dataset into:

- **Training**: 6,000 images (80%)
- **Validation**: 1,000 images (10%)
- **Testing**: 1,000 images (10%)

This dataset served as the ground truth for visual accuracy.

B. Synthetic Style Dataset

A major challenge in stylized captioning is the lack of labeled data; there is no massive dataset of "political" or "romantic" image captions. To mitigate this, we created a **Synthetic Style Dataset**

We curated a list of sentences exemplifying our target styles. This dataset was constructed by taking standard captions and rewriting them using style-specific vocabulary.

- **Funny**: Uses colloquialisms, exaggeration, and humor (e.g., "zooming around," "late for something").
- **Romantic**: Uses flowery, emotional adjectives (e.g., "soft evening," "happily").
- **Academic**: Uses passive voice and formal terminology (e.g., "observed," "canine," "specimen").
- **Political**: Uses terminology related to governance, rights, and public space.

This text-only dataset allowed us to learn the embedding vectors for tokens like $\langle \text{funny} \rangle$ without needing millions of corresponding images.

V. RESULTS AND ANALYSIS

A. Quantitative Metrics

We evaluated the model on the Flickr8k test set. We utilized standard NLP metrics including BLEU (Bilingual Evaluation Understudy) and ROUGE (Recall-Oriented Understudy for Gisting Evaluation). The results are summarized in Table II.

TABLE II
PERFORMANCE METRICS ON TEST SET

Metric	Baseline (Standard)	Ours (Style-Controlled)
BLEU-1	0.45	0.5126
BLEU-4	0.15	0.1529
ROUGE-1	0.40	0.4881
ROUGE-L	0.20	0.1926

Analysis of Quantitative Results:

- 1) **BLEU-1 (0.5126)**: The high unigram overlap score confirms that our model successfully identifies the core objects in the scene (e.g., "dog", "grass", "boy"). This validates that the ResNet encoder and Attention mechanism are functioning correctly.
- 2) **BLEU-4 (0.1529)**: This score measures the overlap of 4-word phrases. Our score is comparable to the baseline. It is important to note that a lower BLEU score is often expected in stylized captioning, as the goal is to *deviate* from the neutral ground truth captions. If the ground truth is "A dog runs," and our model generates "A

majestic canine gallops,” the BLEU score penalizes the model despite the high quality of the output.

- 3) **ROUGE-1 (0.4881)**: This metric indicates a strong recall of key information, suggesting that despite the stylistic changes, the model retains the core informational content of the image.
- 4) **ROUGE-L (0.1926)**: ROUGE-L measures the longest common subsequence between the generated caption and the reference, capturing sentence-level structural similarity. Our ROUGE-L score is slightly lower than the baseline, which is expected in a style-controlled setting where sentence structure and phrasing are intentionally altered to achieve stylistic variation.

B. Qualitative Analysis

The primary success of this project is best demonstrated through qualitative examples, where the model generates distinct descriptions for the same image based on the input style token.



Fig. 2. Qualitative Results: The model generates five distinct captions for the same input image based on the style token provided. Note the shift in vocabulary from "dog" to "canine" in the Academic style.

Case Study: A dog running on grass

- **Style <normal>**: "a dog running through green grass."
Analysis: This represents the baseline performance. The model correctly identifies the subject ("dog"), action ("running"), and setting ("green grass").
- **Style <funny>**: "a happy dog zooming around like it's late for something."

Analysis: The model introduces personality ("happy") and uses a simile ("like it's late for something"). The word "zooming" is a colloquial synonym for running that fits the humorous tone.

- **Style <romantic>**: "a dog running happily through soft evening grass."
Analysis: The model modifies the noun "grass" with "soft" and adds "evening," creating a mood rather than just a description.
- **Style <academic>**: "a canine is observed running across a grassy field."
Analysis: The shift here is syntactic and lexical. "Dog" becomes "canine." The active voice ("dog running") shifts to the passive voice ("is observed"), mimicking the detached tone of scientific observation.
- **Style <political>**: "a stray dog running freely in public space."
Analysis: The model interprets the scene through a sociopolitical lens, using terms like "stray," "freely," and "public space."

VI. CHALLENGES AND MITIGATION STRATEGIES

Throughout the development lifecycle, we encountered several technical challenges.

A. Challenge 1: Generation of Synthetic Style Data

Issue: A key challenge in this work was the generation of high-quality synthetic style data. Since stylized image-caption pairs are scarce, the model relied on text-only synthetic data generated using a Large Language Model (LLM). While this approach enabled scalable style learning, the generated data occasionally contained repetitive phrasing, stylistic inconsistencies, or subtle biases, which could limit the generalization of style control.

Mitigation: To address this, the synthetic dataset was curated to balance different styles and sentence structures. Additionally, style learning was isolated to a second training phase with the visual encoder frozen, reducing the risk of contaminating visual grounding. Future iterations could further improve this process through systematic filtering, diversity constraints, and human validation of the generated samples.

B. Challenge 2: Evaluating Subjective Stylistic Qualities

Issue: Evaluating stylistic attributes such as humor, tone, or expressiveness remains inherently challenging. Standard automatic metrics such as BLEU, ROUGE, and CIDEr focus on n-gram overlap with reference captions and are poorly suited for measuring subjective qualities. As a result, stylistically successful captions may receive low scores if they differ significantly from the reference text, even when they are fluent and contextually appropriate.

Mitigation: To address this limitation, we used a hybrid evaluation strategy. Automatic metrics such as BLEU and ROUGE were primarily used to check basic linguistic fluency and whether the generated captions remained semantically aligned with the image. To evaluate the success of style

transfer, we relied on human qualitative evaluation conducted on a small, randomly selected set of generated captions. This approach allowed us to assess stylistic attributes that cannot be captured by standard automatic metrics.

VII. ETHICAL CONSIDERATIONS

The deployment of controllable image captioning systems raises important ethical considerations, particularly with respect to bias introduced during data generation. In this work, the stylistic datasets were synthetically generated using a Large Language Model (LLM), which may implicitly encode societal, cultural, or political biases present in its training data. As a result, stylistic categories such as `<funny>` may unintentionally include biased humor or stereotypes, which could then be learned and amplified by the captioning model.

Additionally, style categories related to sensitive domains, such as `<political>`, pose a risk of misuse, including the generation of misleading or propagandistic interpretations of visual content. To mitigate these risks, future work should incorporate rigorous dataset curation, automated bias detection, and human-in-the-loop review processes.

VIII. CONCLUSION AND FUTURE WORK

A. Conclusion

We have presented a robust and lightweight architecture for **Style-Controlled Image Captioning** that explicitly separates visual understanding from linguistic expression. By combining a ResNet-50 visual encoder with a style-infused LSTM decoder and an attention mechanism, the proposed model effectively captures salient visual regions while allowing controllable stylistic modulation during caption generation. The introduction of learnable style embeddings injected at every decoding step enables consistent stylistic influence throughout the generation process.

Furthermore, our two-stage training strategy shows that the model can learn different writing styles using only synthetic text data, without the need to retrain or modify the visual encoder. By separating visual learning from style learning, the model is able to adopt new linguistic styles while still accurately describing image content learned from real image-caption pairs.

Overall, this work highlights the feasibility of scalable and flexible style control in image captioning systems, offering a practical foundation for future research in controllable multi-modal generation and human-centered AI applications.

B. Future Work

Future iterations of this project could explore:

- **Beam Search Decoding:** The current system uses greedy decoding, which may lead to repetitive or locally optimal captions. Future work could incorporate Beam Search to explore multiple candidate sequences during inference, potentially improving fluency, coherence, and overall caption quality.
- **LLM-Based Synthetic Style Data Generation:** The effectiveness of style conditioning depends on the quality

of the synthetic style dataset. Large Language Models (LLMs) could be used to generate larger and more diverse stylized text corpora. However, such data may contain biased or harmful content inherited from the LLM’s training data. Future work would focus on carefully filtering and reviewing the generated text to remove biased, stereotypical, or misleading content to ensure that the style data is both reliable and responsibly used.

- **Diffusion-Based Decoders:** The current autoregressive LSTM decoder may limit output diversity. Future research could investigate non-autoregressive diffusion-based decoding methods, such as Prefix-Diffusion, to enhance diversity while reducing exposure bias in sequence generation.

IX. CONTRIBUTIONS

The project was executed with a clear division of labor:

- **Prashanna Mani Paudel:** Lead on the data pipeline. Implemented the `Flickr8kDataset` class, vocabulary builder, and image preprocessing transforms.
- **Toyiomfon Obot Udoete:** Lead on the Baseline Model. Developed the initial CNN-LSTM architecture and validated the ResNet-50 feature extraction.
- **Pratik Shrestha:** Lead on Training and Evaluation. Wrote the training loops, implemented the BLEU/ROUGE evaluation scripts, and tuned the hyperparameters (learning rates, batch sizes).
- **Sanjog Gaihre:** Lead on the Style Extension. Designed the synthetic data generation strategy and modified the `DecoderWithAttention` class to accept and process Style Embeddings.

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